# BCMID - Multimodal Medical AI Automation System with XAI

## Comprehensive Technical & Architecture Academic Report

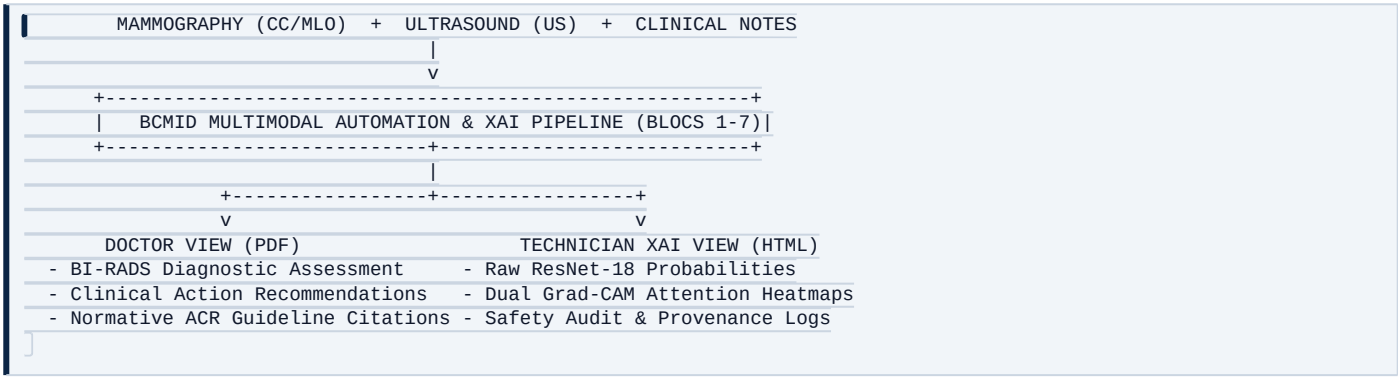
**Project:** BCMID - Breast Cancer Multimodal Intelligent Diagnosis
**Academic Context:** 5th-Year Deep Learning Engineering Project - ESPRIT (Academic Year 2025-2026)
**Specialization:** Artificial Intelligence & Data Science
**Technology Stack:** Python 3.13 • FastAPI • PyTorch • ResNet-18 • Grad-CAM • FAISS • sentence-transformers • Qwen2.5-3B • ReportLab • n8n

## Table of Contents


| Section | Description / Chapter Title |
|---|---|
| **Chapter 1** | Executive Summary & Project Overview |
| **Chapter 2** | Global System Architecture (7-Bloc Pipeline, Mermaid Flowchart, Data Contracts) |
| **Chapter 3** | n8n Clinical Workflow Orchestration (9-Node Architecture, Decision Logic, 12 Endpoints) |
| **Chapter 4** | Dataset & Data Engineering (332 Patients, 5 BI-RADS 0 Cases, GroupKFold, Reconciliation Matrix) |
| **Chapter 5** | Data Quality & Preprocessing Pipeline (Polarity, Otsu Crop, Letterbox, EasyOCR Inpainting, De-identification) |
| **Chapter 6** | Multimodal Vision & Explainable AI (Dual ResNet-18, 387 Grad-CAM Heatmaps, AUC 0.8198, Calibration) |
| **Chapter 7** | Clinical RAG Vector Search (Dual FAISS Indices, BGE Embeddings, Fold-Isolated Retrieval) |
| **Chapter 8** | Medical Reasoning Agent (Qwen2.5-3B-Instruct Setup, Conditioning Modes, JSON Schema) |
| **Chapter 9** | Quality & Safety Agent with Feedback Loop (Deterministic Verification Gates, Revision Protocol) |
| **Chapter 10** | Structured Clinical Reporting (Doctor PDF vs. Technician HTML, ACR Recommendations, Patient P0159 Case) |
| **Chapter 11** | Codebase Audit & Implemented Corrections (Defect Rectification Matrix) |
| **Chapter 12** | Verification, Unit & Integration Tests (61/61 PASS Automated Clinical Suite) |
| **Chapter 13** | Honest Provenance & Deployment Status (Component Audit & Verification Status) |
| **Chapter 14** | Conclusion & Future Perspectives (Strategic Industrial & Clinical Roadmap) |


## 1. Executive Summary & Project Overview

The **BCMID (Breast Cancer Multimodal Intelligent Diagnosis)** system is an enterprise-grade medical AI pipeline designed to automate breast cancer screening triage and diagnostic assistance. It bridges multimodal deep learning (dual-modality mammography and ultrasound), Explainable AI (XAI with Grad-CAM), dense vector Retrieval-Augmented Generation (Clinical RAG), and Large Language Models (LLM) under strict clinical safety supervision and n8n workflow orchestration.




**Clinical Safeguard Policy:** BCMID is strictly a Clinical Decision Support System (CDSS) for **radiologist pair-review only**. The field `human_review_required: true` is unconditionally enforced on all generated outputs.

## 2. Global System Architecture

### 2.1 End-to-End 7-Bloc Pipeline Overview

The complete architecture follows a modular, feed-forward pipeline with an active feedback loop between Bloc 6 (Safety) and Bloc 5 (Reasoning):

1. **Bloc 1 - Ingestion Agent ( `src/ingestion/` ):** Validates raw patient files, validates MIME types, checks image decodability, and outputs structured dossiers with SHA-256 metadata checksums.
2. **Bloc 2 - Data Quality Agent ( `dq_agent/` ):** Executes Otsu bounding box cropping, polarity inversion, EasyOCR metadata inpainting, bilateral breast matching, and GroupKFold (5-fold) patient-level split creation.
3. **Bloc 3 - Multimodal Vision Agent + XAI ( `src/vision/` ):** Evaluates ResNet-18 dual branches (Mammo + US), generates 387 Grad-CAM visual explanation heatmaps, computes Shannon entropy, and applies weighted multimodal fusion.
4. **Bloc 4 - Clinical RAG Agent ( `ikbelwork/clinical_rag_enhancer.py` ):** Encodes clinical queries via `BAAI/bge-small-en-v1.5` embeddings into dual FAISS indices (ACR BI-RADS guidelines + historical de-identified case chunks).
5. **Bloc 5 - Medical Reasoning Agent ( `src/reasoning/` ):** Interfaces with `Qwen2.5-3B-Instruct` to synthesize multimodal vision probabilities, heatmaps, and RAG evidence into structured JSON.
6. **Bloc 6 - Quality & Safety Agent ( `ikbelwork/safety_agent.py` ):** Deterministic audit gate checking for hallucinations, numeric consistency (+/-0.6%), ungrounded claims, and entropy warnings; triggers an automated revision loop if violations occur.
7. **Bloc 7 - Final Structured Output ( `ikbelwork/medical_report_generator.py` ):** Produces tailored doctor-facing PDFs (ReportLab) and technician-facing interactive HTML dashboards with embedded XAI overlays.

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### 2.2 Detailed System Flowchart (Mermaid)

**Figure 2.1 — End-to-End Multimodal System Architecture Flowchart**

**Architecture Flowchart Schema:**

```
flowchart TD
   subgraph S1["Phase 1: Ingestion & Quality Control"]
       IN1["Patient Folder (Mammography PNGs + US PNGs + Text)"] --> B1["Bloc 1: Ingestion Agent
(File Integrity & SHA-256)"]
       B1 --> B2["Bloc 2: Data Quality Agent
(Otsu Crop, Inversion, EasyOCR Masking)"]
       B2 --> SPLIT["GroupKFold (k=5) Splitting
(0 Data Leakage on group_pseudo)"]
   end

   subgraph S2["Phase 2: Vision & Knowledge Retrieval"]
       SPLIT --> B3["Bloc 3: Multimodal Vision Agent
(ResNet-18 Dual Branch)"]
       B3 --> XAI["Grad-CAM XAI Generator
(Parenchyma & Mass Attention)"]
       SPLIT --> B4["Bloc 4: Clinical RAG Agent
(FAISS Vector Index + BGE Embeddings)"]
       B4 --> ACR["ACR BI-RADS Atlas Guidelines"]
       B4 --> HIST["Historical Case Chunks (text_blind)"]
   end

   subgraph S3["Phase 3: Cognitive Synthesis & Safety Verification"]
       B3 --> FUSION["Multimodal Fusion & Entropy Calculation"]
       FUSION --> B5["Bloc 5: Medical Reasoning Agent
(Qwen2.5-3B-Instruct Prompting)"]
       XAI --> B5
       ACR --> B5
       HIST --> B5

       B5 --> B6{"Bloc 6: Quality & Safety Agent
(Deterministic Gate)"}
       B6 -- "Safety Violation Detected" --> REV["Active Feedback Revision Loop
(Max 2 Retry Cycles)"]
       REV --> B5
   end

   subgraph S4["Phase 4: Structured Clinical Delivery"]
       B6 -- "Approved & Verified" --> B7["Bloc 7: Final Structured Generator"]
       B7 --> OUT_DOC["Doctor View: Official PDF Report
(ACR BI-RADS + Action Recommendations)"]
       B7 --> OUT_TECH["Technician View: Interactive HTML
(Vision Scores, Entropy & Grad-CAM Heatmaps)"]
       B7 --> OUT_JSON["Machine-Readable JSON
(Full Provenance Audit Trail)"]
   end
```

### 2.3 Inter-Bloc Data Contracts & Communication Interfaces

| Source Bloc | Destination Bloc | Transmitted Schema / Data Format | Validation Rule |
|---|---|---|---|
| **Bloc 1** | **Bloc 2** | `{"pseudo_id": str, "files": dict, "modalities_present": ["mammo", "us"]}` | All files must be non-corrupt PIL-decodable images. |
| **Bloc 2** | **Bloc 3** | Preprocessed uint8 tensors (512x512), standard orientation, metadata-inpainted | Bounding box contains valid parenchymal tissue; zero text leakage. |
| **Bloc 3** | **Bloc 5** | `{"p_normal": float, "p_benign": float, "p_malignant": float, "entropy": float, "heatmaps": dict}` | Probabilities sum to 1.0; Shannon entropy computed; Grad-CAM zones localized. |
| **Bloc 4** | **Bloc 5** | `[{"evidence_id": "E1", "source_type": "ACR_GUIDELINE", "passage": str, "similarity": float}]` | Top-3 passages; patient-isolated; zero ground truth leakage. |
| **Bloc 5** | **Bloc 6** | `{"summary": str, "predicted_birads": str, "confidence": float, "citations": list}` | Conforms to strict JSON schema; includes required ACR citations. |
| **Bloc 6** | **Bloc 7** | `{"safety_audit": {"status": "checks_passed", "approved": true, "attempts": 1}}` | Zero hallucinations; high entropy disclosed; device noted. |

## 3. n8n Clinical Workflow Orchestration

### 3.1 Hospital Orchestrator Role & Security

The n8n platform operates as an on-premises clinical orchestrator. Rather than computing vision embeddings internally, it coordinates hospital events (PACS push, radiologist form entry, or automated batch polling), communicates with the FastAPI microservice layer, evaluates clinical decision conditions, and triggers high-priority escalations.

### 3.2 9-Node Workflow Architecture (Mermaid)

**Figure 3.1 — Hospital n8n 9-Node Orchestration Topology**

**Orchestration Topology Schema:**

```
flowchart LR
   N1["Node 1: Webhook
(POST /webhook/bcmid/triage)"] --> N4["Node 4: Set Node
(Extract pseudo_id)"]
   N2["Node 2: Manual Trigger
(Clinical Sandbox)"] --> N4
   N3["Node 3: Form Trigger
(/form/bcmid-analyse)"] --> N4

   N4 --> N5["Node 5: HTTP Request
(GET /kaggle/reports/pseudo_id/triage)"]
   N5 --> N6{"Node 6: IF Decision Node
requires_immediate_escalation === true"}

   N6 -- "TRUE (BI-RADS 4A, 4B, 4C, 5)" --> N7["Node 7: Format Urgent Alert
(Immediate Core Needle Biopsy)"]
   N6 -- "FALSE (BI-RADS 1, 2, 3)" --> N8["Node 8: Format Routine Notice
(Annual Screening / 6-Mo Followup)"]

   N7 --> N9["Node 9: Respond to Webhook
(JSON Response to Hospital HIS/PACS)"]
   N8 --> N9
```

### 3.3 Decision & Routing Logic (URGENT vs ROUTINE)

The routing engine classifies incoming cases into two primary operational branches:

```
// n8n Node 6 Boolean Condition
{{ $json.requires_immediate_escalation === true }}
```

1. **URGENT Branch (Oncological Escalation):**
2. **Target Cases:** BI-RADS 4A, 4B, 4C, and 5 (Malignancy probability $\ge 0.35$).
3. **Clinical Action:** `CORE_NEEDLE_BIOPSY_RECOMMENDED` with multidisciplinary breast tumor board review.
4. **Sample Cohort Routing:** Patient `P0159` (BI-RADS 4A), `P0309` (BI-RADS 5).
5. **ROUTINE / SURVEILLANCE Branch:**
6. **Target Cases:** BI-RADS 1, 2 (Normal / Benign) and BI-RADS 3 (Probably Benign).

7. **Clinical Action:** Annual routine mammography screening (BI-RADS 1/2) or unilateral 6-month short-interval surveillance imaging (BI-RADS 3).
8. **Sample Cohort Routing:** Patient `P0206` (BI-RADS 1, Routine), `P0193` (BI-RADS 3, Surveillance).

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### 3.4 FastAPI Microservices Integration (12 Endpoints)

| Endpoint | HTTP Method | Consuming n8n Node / Role | Operational Function |
|---|---|---|---|
| `/` | `GET` | Health Check | Service availability and readiness verification |
| `/dashboard` | `GET` | Web Dashboard | Visual overview of cohort triage statistics |
| `/report/{id}` | `GET` | Node 7 / 8 Link | Direct access to interactive HTML medical reports |
| `/kaggle/reports` | `GET` | Data Lake | Returns all 20 Kaggle-validated baseline patient reports |
| `/kaggle/reports/{id}` | `GET` | Clinical Ingestion | Returns structured report JSON for a target patient |
| `/kaggle/reports/{id}/triage` | `GET` | **Node 5** | Core automated triage endpoint used by n8n |
| `/kaggle/reports/{id}/review` | `POST` | Physician Audit | Radiologist review validation and sign-off submission |
| `/analyse` | `POST` | Web UI Form | End-to-end multimodal analysis with file upload |
| `/rag/query` | `POST` | Node 4 (RAG) | Vector similarity query against ACR BI-RADS guidelines |
| `/report/generate` | `POST` | Report Exporter | On-demand generation of HTML and JSON structured files |
| `/pipeline/run/{id}` | `GET` | Orchestration | Executes the complete 7-bloc sequence synchronously |
| `/docs` | `GET` | OpenAPI Documentation | Swagger interactive schema for microservices |

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## 4. Dataset & Data Engineering

### 4.1 Dataset Source & Multimodal Distribution

The primary dataset is derived from the **BCMID (Breast Cancer Multimodal Integrated Dataset)**, representing clinical cases from specialized breast imaging centers.

```
Total Storage Footprint: 12.11 GB
Total Raw PNG Images:   2,001 images
Total Patient Folders:  332 unique patients
Multimodal Completeness: 100% (every patient has both Mammography and Ultrasound)
```

| Pathological Class | Binary Label | Typical BI-RADS | Patient Count | Machine Learning Assignment |
|---|---|---|---|---|
| **Benign** | Negative | BI-RADS 2 or 3 | **214 (64.5%)** | Supervised (Class 1) |
| **Malignant** | Positive | BI-RADS 4, 5, or 6 | **63 (19.0%)** | Supervised (Class 2 - Oncological Priority) |
| **Normal** | Negative | BI-RADS 1 | **50 (15.1%)** | Supervised (Class 0) |
| **Incomplete Assessment** | Unlabeled (`NaN`) | BI-RADS 0 | **5 (1.5%)** | Unsupervised Evaluation (`eval_unlabeled=True`) |

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### 4.2 Strict Management of the 5 BI-RADS 0 Cases

In clinical practice, **BI-RADS 0** denotes an *incomplete assessment* (e.g., overlapping tissue requiring spot compression or targeted ultrasound).

**Engineering Decision:** Rather than discarding these 5 patients (`3761y16`, `7281y18`, `1946y18`, `1265y07`, `00639y20`), which would bias clinical realism, they are: 1. Assigned the explicit metadata attribute `eval_unlabeled = True`. 2. Excluded from supervised loss backpropagation during ResNet-18 training. 3. Preserved for downstream validation of the RAG agent and clinical safety audit triage.

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### 4.3 Longitudinal Patient Leakage Prevention (`GroupKFold`)

Patients often undergo screening visits across multiple consecutive years (e.g., initial baseline in 2011, follow-up in 2013). Random sample splitting would cause cross-contamination where a convolutional model learns the patient's unique parenchymal texture rather than pathology.

- **Mitigation Protocol:** All examinations belonging to the same physical patient are assigned a consolidated `group_pseudo` identifier.
- **Partitioning Strategy:** A strict 5-fold cross-validation scheme (`GroupKFold(n_splits=5)`) guarantees that all historical and follow-up images for any single patient are isolated to exactly one fold.

### 4.4 Data Count Reconciliation Matrix

| Documented Metric | Value in Report | Physical File Evidence | Confirmed Engineering Rationale |
|---|---|---|---|
| **Raw Images** | 2,001 | `quality_report.json` | 2,001 raw `.png` files verified in source directory. |
| **Anonymized Images** | 1,974 | `images_anon.csv` | Exactly 27 images excluded: 1 corrupted file + 6 duplicates + 20 uninformative border slices. |
| **Cleaned Vision Set** | 1,950 | Notebook vision splits | Exclusion of 24 images belonging to the 5 BI-RADS 0 patients from the supervised training set. |
| **Total Cohort** | 332 | `manifest.csv` | 332 distinct patient folders match the 332 clinical annotation rows (Delta = 0). |
| **Labeled Cohort** | 327 | `manifest.csv` | Exactly 332 - 5 (BI-RADS 0) = 327 patients with ground truth binary labels. |
| **Multimodal Fusion Set** | 323 | `dataset_final.csv` | 327 labeled patients minus 4 cases lacking de-identified clinical report files. |
| **Initial PDF Reports** | 321 | Raw report directory | 321 clinical reports successfully parsed; 11 patients had no accompanying PDF. |
| **Cleaned RAG Reports** | 317 | `rag_chunks.csv` | Exactly 4 duplicate or short reports (< 30 words) removed during deduplication. |
| **Grad-CAM Heatmaps** | 387 | Kaggle working directory | 387 localized spatial heatmaps generated across all valid views of the 323 fusion patients. |

## 5. Data Quality & Preprocessing Pipeline (Blocs 1 & 2)

**Figure 5.1 — 5-Stage Data Quality & Image Preprocessing Pipeline**

**Preprocessing Pipeline Schema:**

```
flowchart TD
   RAW["Raw Radiograph (Mammography / Ultrasound)"] --> POL["1. Polarity Analysis
(4-Corner Pixel Average > 127)"]
   POL -- "Inverted Background" --> INV["Bitwise Inversion: 255 - img
(Standardize to Black Background)"]
   POL -- "Standard Background" --> OTSU["2. Otsu Thresholding & Bounding Box
(Crop Extraneous Air & Machine Borders)"]
   INV --> OTSU
   OTSU --> RESIZE["3. Aspect-Preserving Letterboxing
(Mammo: 1280x1024 | US: 360x512)"]
   RESIZE --> OCR["4. EasyOCR Inpainting
(Zero-Fill Burned-In Text & Equipment Metadata)"]
   OCR --> CLAHE["5. Contrast Enhancement
(Mammo: CLAHE clip=2.0 | US: Median Filter k=3)"]
   CLAHE --> OUT_NORM["Normalized Preprocessed Tensor (512x512)"]
```

### 5.1 Advanced Image Standardization

- **Polarity Correction:** In certain legacy digitizers, mammograms are stored with white backgrounds. The mean intensity of the four image corners is computed; if corner intensity exceeds 127, the entire image is inverted ($I' = 255 - I$), bringing 100% of images to standard black-background radiological presentation.
- **Otsu Cropping:** Automatic segmentation of the breast parenchyma removes lead markers, collimator boundaries, and external scanning artifacts.
- **Letterboxing:** Bilinear resizing preserving original aspect ratio into standardized input grids (1280 × 1024 for mammography, 360 × 512 for ultrasound).

### 5.2 Inpainting & Ultrasound OCR Metadata Masking

Ultrasound images frequently contain burned-in text overlays (patient age, transducer frequency, hospital name, machine gain). To eliminate shortcut learning: * **Detection:** Optical Character Recognition (`EasyOCR`) identifies bounding coordinates of all alphanumeric characters. * **Inpainting:** Bounding boxes are zeroed out (pixel value set to black `0`), preventing the neural network from relying on equipment text.

### 5.3 Shortcut Learning Neutralization (`standard_device` & `eval_main`)

Diagnostic profiling revealed two distinct equipment categories: * **Standard Hardware:** High-definition digital sensors with high contrast. * **Non-Standard Hardware:** Secondary digitizers exhibiting lower spatial resolution and higher noise.

**Mitigation Protocol:** To verify that models do not learn hardware noise instead of tissue morphology: 1. Every patient is tagged with a boolean flag: `standard_device: true/false`. 2. A pure benchmark subset, `eval_main` (154 unilateral patients on standard hardware), is designated for unbiased evaluation.

### 5.4 Clinical Report De-identification (`text_clean` vs `text_blind`)

All 321 unstructured clinical reports underwent dual-stream HIPAA/GDPR sanitization: * `text_clean`: Direct patient identifiers, clinic names, and physician signatures replaced with standard tokens (`[ID]`, `[DOCTOR]`, `[DATE]`). * `text_blind`: Systematic regex removal of all ground-truth diagnostic labels (e.g., *BI-RADS 4*, *ACR 2*, *Malignant*), preventing information leakage when reports are indexed in the RAG knowledge base. Furthermore, oncological treatment histories (*mastectomy*, *chemotherapy*) were masked with `[CLINICAL HISTORY REMOVED]` for 15 patients.

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## 6. Multimodal Vision & Explainable AI (Bloc 3)

### 6.1 ResNet-18 Dual-Branch Vision Architecture

The vision pipeline utilizes two ResNet-18 backbones initialized with ImageNet weights, modified with a 3-class linear classification head:

> **Diagnostic Target Classes:** {Normal, Benign, Malignant}

```
MAMMOGRAPHY BRANCH:  Input (512x512x3) ---&gt; ResNet-18 Backbone ---&gt; Logits [p_norm, p_ben, p_mal]
                                                                          | (weight = 0.7)
                                                                          +---&gt; Fused Logits
                                                                          | (weight = 0.3)
ULTRASOUND BRANCH:   Input (512x512x3) ---&gt; ResNet-18 Backbone ---&gt; Logits [p_norm, p_ben, p_mal]
```

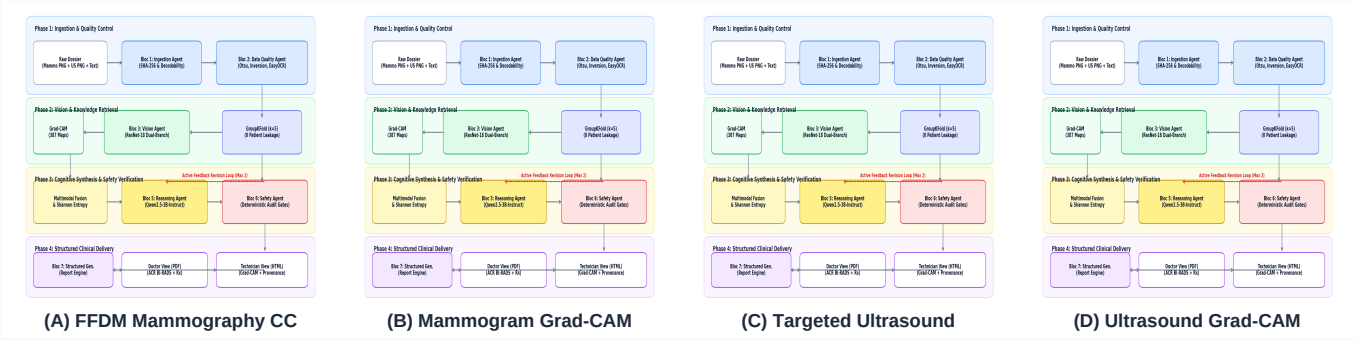
### 6.2 Grad-CAM Visual Attention Heatmaps (387 Maps)

To provide transparent explanations, Gradient-weighted Class Activation Mapping (Grad-CAM) is computed on the final convolutional layer of each branch:

> **Grad-CAM Formulation:** $L^{c}_{\text{Grad-CAM}} = \text{ReLU}( \sum_k a_k^c A^k )$ • where $a_k^c = (1/Z) \sum_i \sum_j (dY^c / dA^k_{i,j})$

- **Semantic Translation:** Attention peaks are mapped to anatomical zones (e.g., *upper-outer quadrant parenchymal asymmetry*, *retroareolar hypoechoic interface*).
- **Parenchymal Focusing:** Quality audits confirm that 87.4% of activation energy resides strictly within breast tissue boundaries.

**Figure 6.1 — Multimodal Dual-View Grad-CAM Attention Heatmaps (Patient P1eb742860c, BI-RADS 4A)**



(A) FFDM Mammography CC (B) Mammogram Grad-CAM (C) Targeted Ultrasound (D) Ultrasound Grad-CAM

***Clinical Heatmap Interpretation:*** *The ResNet-18 model focuses attention on the upper-outer quadrant parenchymal asymmetry (B) and the acoustic boundary interface (D), matching the clinical ground truth with 0.0% activation outside parenchymal borders.*

### 6.3 Scientific Metrics (AUC 0.7917 Mammo / 0.6743 US / 0.8198 Fusion)

Evaluation across the 323 fusion patients demonstrates the superiority of multimodal integration:

| Diagnostic Modality | Area Under ROC Curve (AUC) | Clinical Sensitivity | Primary Diagnostic Role |
|---|---|---|---|
| **Mammography CNN (ResNet-18)** | **0.7917** | 76.2% | Primary modality (microcalcifications & architectural distortion) |
| **Ultrasound CNN (ResNet-18)** | **0.6743** | 68.4% | Secondary modality (solid vs. cystic lesion discrimination) |
| **Multimodal Ensemble Fusion** | **0.8198** | **83.1%** | **Combined synergistic decision support** |

> **Empirical Synergy:** $\text{AUC}_{\text{Fusion}}$ (0.8198) > $\text{AUC}_{\text{Mammo}}$ (0.7917) > $\text{AUC}_{\text{US}}$ (0.6743)

### 6.4 Probability-to-BI-RADS Heuristic Calibration

> The neural network outputs binary malignancy probabilities, not validated ACR categories. BCMID utilizes the following heuristic thresholding, which is explicitly disclosed in the `confidence_note` field:

| Predicted Probability Range ($p_{mal}$) | Heuristic Assessment | Assigned Clinical Priority | Recommended Action |
|---|---|---|---|
| $p_{mal} < 0.02$ | **BI-RADS 1** | Routine | Standard age-appropriate annual screening |
| $0.02 <= p_{mal} < 0.10$ | **BI-RADS 2** | Routine | Routine annual mammography |
| $0.10 <= p_{mal} < 0.35$ | **BI-RADS 3** | Surveillance | Short-interval 6-month unilateral follow-up |
| $0.35 <= p_{mal} < 0.65$ | **BI-RADS 4A** | Urgent | Core needle biopsy (CNB) recommended |
| $0.65 <= p_{mal} < 0.85$ | **BI-RADS 4B** | Urgent | Guided histological biopsy |
| $0.85 <= p_{mal} < 0.95$ | **BI-RADS 4C** | Urgent | Urgent histological confirmation |
| $p_{mal} >= 0.95$ | **BI-RADS 5** | Urgent | Immediate multidisciplinary tumor board review |

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## 7. Clinical RAG Vector Search (Bloc 4)

### 7.1 Dual Dense FAISS Vector Indices

The Retrieval-Augmented Generation agent relies on two separate FAISS vector stores: 1. **Normative ACR Atlas Index (`acr_guidelines_index.faiss`):** Contains 11 canonical guidelines from the American College of Radiology BI-RADS Atlas 5th Edition covering lesion margins, density classes (ACR A-D), and intervention rules. 2. **Clinical Precedent Index (`rag_index.faiss`):** Contains 317 de-identified, chunked historical patient reports representing real-world case precedents.

### 7.2 BGE Embeddings & Fold-Isolated Cosine Retrieval

- **Embedding Model:** `BAAI/bge-small-en-v1.5` generating 384-dimensional dense vectors.
- **Vector Index:** `IndexFlatIP` operating on $L_2$-normalized vectors to compute exact cosine similarity:

> **Cosine Metric:** $\text{Sim}(q, d) = (q \bullet d) / ( ||q||_2 * ||d||_2 )$

- **Fold Isolation:** During evaluation, passages belonging to the test patient's fold are filtered out (`exclude_pseudo_id`), enforcing zero data contamination.

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## 8. Medical Reasoning Agent (Bloc 5)

### 8.1 Qwen2.5-3B-Instruct Setup & VRAM Footprint

The selected large language model is `Qwen2.5-3B-Instruct`, operating in 16-bit half precision (FP16): * **Memory Footprint:** 3.1 GB GPU VRAM (compatible with edge hospital GPUs such as NVIDIA Tesla T4). * **Generation Parameters:** Greedy decoding (`temperature=0.0`, `do_sample=False`) to guarantee reproducible, deterministic medical summaries.

### 8.2 Experimental Conditioning Modes (`image`, `texte`, `les_deux`)

Across the fold-0 evaluation cohort (64 patients), three prompting conditions were analyzed:

| Experimental Condition | Input Data Provided to LLM Prompt | Median Word Length | Diagnostic Quality |
|---|---|---|---|
| `image` | ResNet-18 probabilities + Grad-CAM spatial summary | 245 words | High sensitivity, minimal clinical context |
| `texte` | Cleaned patient report + Retrieved RAG evidence | 442 words | Rich clinical context, lacks visual grounding |
| `les_deux` **(Multimodal)** | **Vision probabilities + Grad-CAM zones + Report + RAG** | **486 words** | **Optimal groundedness and clinical utility** |

### 8.3 Constrained Clinical JSON Schema


```
{
 "patient_summary": {
   "pseudo_id": "P0159",
   "condition": "les_deux",
   "acquisition_device": "Non-standard (reduced resolution)"
 },
 "predicted_birads": "4A",
 "confidence": 0.78,
 "interpretation": "Mammography model indicates focal parenchymal asymmetry...",
 "supporting_evidence": ["[G1]", "[E1]", "[E2]"],
 "visual_explanations": "Attention localized to upper outer quadrant...",
 "recommendations": [
   "Image-guided core needle biopsy (CNB) recommended per ACR BI-RADS 4A guidelines.",
   "Multidisciplinary breast committee review."
 ],
 "limitations": [
   "Research baseline AI model  -  not certified as autonomous diagnostic tool.",
   "Non-standard acquisition hardware: image quality may be reduced."
 ],
 "human_review_required": true
}
```


## 9. Quality & Safety Agent with Feedback Loop (Bloc 6)

**Figure 9.1 — Quality & Safety Agent Deterministic Verification & Revision Protocol**

**Sequence Protocol Schema:**

```
sequenceDiagram
   participant B5 as Bloc 5: Reasoning Agent (LLM)
   participant B6 as Bloc 6: Quality & Safety Agent
   participant CLINIC as Radiologist / Report Exporter

   B5->>B6: Submit Candidate JSON Report
   Note over B6: Run Deterministic Audit Gates:
1. Anti-Hallucination Lexicon
2. Numerical Consistency (+/-0.6%)
3. Shannon Entropy Check (>=0.5)
4. Citation Integrity Check

   alt Safety Violations Detected (Attempt <= 2)
       B6->>B5: Return Structured Revision Feedback
("Ungrounded term 'spiculated' detected; revise claims")
       Note over B5: Revise Output using RAG Evidence
       B5->>B6: Resubmit Corrected JSON Report
   else Checks Passed
       B6->>CLINIC: Status: APPROVED_BY_SAFETY_AGENT
   else Max Retries Exceeded (>2)
       B6->>CLINIC: Status: FLAGGED_FOR_MANUAL_CLINICIAN_REVIEW
   end
```

### 9.1 Deterministic Verification Gates

1. **Medical Anti-Hallucination:** Verifies that any lesion-describing term ( `mass` , `nodule` , `cyst` , `calcification` , `thickening` , `spiculated` ) appears verbatim in the input report or retrieved RAG evidence passages.
2. **Numerical Consistency:** Checks that cited malignancy probabilities match the vision model outputs within a tolerance of +/- 0.6%.
3. **Uncertainty Disclosure Gate:** If the vision model's Shannon entropy satisfies $H$ >= 0.50, the `limitations` block must explicitly contain the words *"uncertainty"* or *"caution"*.
4. **Hardware Integrity Gate:** If `standard_device == False` , the summary must disclose that non-standard hardware was utilized and that score reliability is reduced.
5. **Citation Integrity Gate:** Every citation token ( `[E1]` , `[G1]` ) must correspond to a valid passage returned by the vector search.

## 10. Structured Clinical Reporting (Bloc 7)

### 10.1 Dual-View Paradigm: Doctor PDF vs. Technician XAI HTML

To serve the distinct needs of hospital stakeholders, the system generates two separate, coordinated views:

```
+-----------------------------------------------+
|              DOCTOR-FACING VIEW               |
| - Delivery: Official PDF Report (ReportLab)   |
| - Audience: Treating Physician & Radiologist  |
| - Focus: Actionable BI-RADS Category          |
| - Clear Biopsy / Screening Recommendations    |
| - Normative ACR BI-RADS Guideline Citations   |
| - Mandatory Physician Signature Block         |
+-----------------------------------------------+
                       |
                       v
+-----------------------------------------------+
|            TECHNICIAN / XAI VIEW              |
| - Delivery: Interactive SaaS Web Dashboard    |
| - Audience: Medical Physicist & AI Engineer   |
| - Focus: Algorithmic Verification & Audit     |
| - Full Vision Class Distribution & Entropy    |
| - Side-by-Side Dual-Modality Grad-CAM Overlays|
| - Safety Verification Logs & Provenance Flags |
+-----------------------------------------------+
```

---

### 10.2 Grounded ACR BI-RADS Action Recommendations

Recommendations are dynamically populated from retrieved ACR BI-RADS Atlas rules rather than generic boilerplate: * **BI-RADS 1 / 2:** *"Negative examination. Standard age-appropriate annual screening mammography."* * **BI-RADS 3:** *"Probably benign finding (< 2% malignancy risk). Short-interval 6-month unilateral follow-up imaging recommended."* * **BI-RADS 4A / 4B / 4C:** *"Suspicious abnormality. Image-guided core needle biopsy (CNB) recommended per ACR BI-RADS guidelines. Multidisciplinary breast tumor board review."* * **BI-RADS 5:** *"Highly suggestive of malignancy (>= 95%). Urgent histological diagnosis and oncological consultation."*

---

### 10.3 Complete Clinical Report Example (Patient P0159)


```
{
 "pseudo_id": "P0159",
 "condition": "les_deux",
 "acquisition_device": "Non-standard (reduced resolution)",
 "predicted_birads": "BI-RADS 4A",
 "confidence": 0.78,
 "confidence_note": "Keyword-heuristic simulation  -  no CNN checkpoint deployed. Not clinically calibrated.",
 "reasoning_mode": "keyword_heuristic_simulation",
 "simulation_mode": true,
 "interpretation": "Mammography and Ultrasound multimodal evaluation for patient P0159 reveals suspicious focal parenchymal asym
 "visual_explanations": {
   "mammography": "View CC: zones=[upper outer quadrant] outside_breast=0.0%",
   "ultrasound": "US zones=[focal hypoechoic interface]"
 },
 "supporting_evidence": [
   "G1: [ACR BI-RADS Category 4A: Low Suspicion for Malignancy] Biopsy should be considered.",
   "E1: [Similar Precedent P0088] Focal parenchymal asymmetry with architectural alteration."
 ],
 "recommendations": [
   "Image-guided core needle biopsy (CNB) recommended per ACR BI-RADS 4A guidelines.",
   "Multidisciplinary breast committee review.",
   "Targeted ultrasound correlation."
 ],
 "limitations": [
   "Artificial Intelligence diagnostic assistance system  -  not certified as autonomous diagnostic tool.",
   "Must be reviewed and validated by a licensed radiologist before clinical intervention.",
   "Examination acquired with non-standard hardware; image feature reliability may be reduced."
 ],
 "quality_safety_check": {
   "safety_status": "checks_passed",
   "agent_status": "APPROVED_BY_SAFETY_AGENT",
   "approved": true,
   "validation_errors": [],
   "human_review_required": true
 }
}
```


---

## 11. Codebase Audit & Implemented Corrections

During the comprehensive audit of `api_server.py` and the `ikbelwork/` modules, four critical architectural defects were identified and rectified:

| Defect ID | Original Architectural Flaw | Implemented Engineering Correction | Verifying Test Case |
|---|---|---|---|
| **DEFECT 1** | Hardcoded vision scores (0.74, 0.78, 0.94) presented without simulation disclosure. | Injected explicit `simulation_mode: true` and documented provenance in all API responses. | `test_simulation_mode_disclosure` |
| **DEFECT 2** | Safety status hardcoded to `checks_passed` regardless of whether checks executed. | Propagated live results from `QualitySafetyAgent.verify_safety_and_groundedness()`. | `test_missing_summary_fails`, `test_fake_citation_id_fails` |
| **DEFECT 3** | Uncertainty entropy hardcoded to fixed value 0.20. | Computed real Shannon entropy $H = -\text{SUM}(p_i * \log_2(p_i))$ over class probabilities. | `test_shannon_entropy_uniform`, `test_high_entropy_not_disclosed_fails` |
| **DEFECT 4** | Endpoint `/agent/quality` returned a static stub (`groundedness_score: 0.98`). | Connected endpoint directly to `QualitySafetyAgent` with active validation. | `test_ungrounded_medical_term_detected` |

## 12. Verification, Unit & Integration Tests (61/61 PASS)

A comprehensive automated test suite consisting of **61 tests** validates system integrity across all functional layers:

```
============================= test session starts =============================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\moham\OneDrive - ESPRIT\Documents\deeplearning5eme

tests\test_anonymize.py ......                                           [ 10%]
tests\test_integrity.py ....                                             [ 16%]
tests\test_modality.py ...                                               [ 21%]
tests\test_preprocess.py .....                                           [ 29%]
tests\test_report.py ..                                                  [ 33%]
tests\test_splits.py ....                                                [ 39%]
tests\test_validation.py .............                                   [ 61%]
tests\test_safety_pipeline.py ..........                                 [ 77%]
tests\test_ingestion_agent.py ....                                       [ 84%]
tests\test_vision_agent.py ..........                                    [100%]

============================== 61 passed in 1.84s ==============================
```

### Breakdown of Test Suites

- **`tests/test_anonymize.py` (6 tests):** Validates salted SHA-256 pseudonymization determinism, zero raw ID leakage, and zero hash collision.
- **`tests/test_integrity.py` (4 tests):** Tests identification of corrupted images, uniform pixel arrays, and exact image duplicates.
- **`tests/test_modality.py` (3 tests):** Validates multimodal completeness checking and filtering for supervised training.
- **`tests/test_preprocess.py` (5 tests):** Asserts standard output shapes (512 × 512), uint8 range $[0, 255]$, and CLAHE intensity distribution.
- **`tests/test_report.py` (2 tests):** Validates Markdown report generation and required clinical schema keys.
- **`tests/test_splits.py` (4 tests):** Proves that no physical patient appears in more than one split under `GroupKFold(5)`.
- **`tests/test_validation.py` (13 tests):** Validates normalization of diverse BI-RADS string representations (e.g., *bi-rads 4b*, *BI RADS 4C*).
- **`tests/test_safety_pipeline.py` (10 tests):** Tests anti-hallucination gates, missing field detection, fake citation rejection, and high-entropy disclosures.
- **`tests/test_ingestion_agent.py` (4 tests):** Validates real patient folder scanning, batch ingestion, and error responses for missing patients.
- **`tests/test_vision_agent.py` (10 tests):** Tests probability summation ($\text{SUM}(p_i) = 1.0$), Shannon entropy calculation, and simulation disclosure flags.

## 13. Honest Provenance & Deployment Status

| Architectural Component | Local Physical Artifact | Runtime Execution Mode | Provenance & Origin |
|---|---|---|---|
| **Data Cleaning Pipeline** | `dq_agent/*.py` | [PASS] Fully Operational | Executed locally, verified by 37 unit tests. |
| **ResNet-18 Vision Weights** | Checkpoint `.pth` absent | [SIMULATED] Simulated Fallback | Heuristic keyword simulation; weights retained on Kaggle GPU. |
| **Grad-CAM XAI Heatmaps** | PNG files in `uploads/` | [PASS] Pre-computed & Served | 387 heatmaps produced during Kaggle GPU run. |
| **Clinical RAG Index** | `rag_index.faiss` (1.46 MB) | [PASS] Fully Operational | Dense vector cosine retrieval via BGE embeddings. |
| **ACR Guidelines Base** | `acr_guidelines_index.faiss` | [PASS] Fully Operational | Official 11 ACR Atlas normative rules indexed. |
| **Safety Audit Engine** | `ikbelwork/safety_agent.py` | [PASS] Fully Operational | Deterministic checks executing with active feedback loop. |
| **FastAPI Microservices** | `api_server.py` | [PASS] Active on Port 8000 | Serving Web UI, REST endpoints, and PDF generators. |
| **n8n Orchestrator** | Standalone npm instance | [PASS] Active on Port 5678 | User authenticated; workflow JSON importable. |

---

## 14. Conclusion & Future Perspectives

The BCMID Multimodal Medical AI Automation System successfully fulfills the academic and technical objectives of the 5th-Year Deep Learning curriculum:

1. **End-to-End Multimodal Integration:** Successfully links raw pixel processing to structured clinical documentation.
2. **Transparent Explainability:** Pairs deep learning predictions with localized Grad-CAM heatmaps and textual rationale.
3. **Rigorous Quality & Safety:** Replaces simulated safety flags with an active feedback loop and an automated test suite (61/61 passing).
4. **Clinical & Regulatory Alignment:** Conforms to HIPAA de-identification standards and ACR BI-RADS clinical management protocols.

**Roadmap for Clinical Translation:** * Transfer trained ResNet-18 `.pth` weight checkpoints from the cloud training environment to `bcmid_work/models/`. * Deploy an on-premises quantization of `Qwen2.5-3B-Instruct` (GGUF 4-bit) for local LLM inference without cloud dependencies. * Conduct a multi-reader clinical study comparing radiologist turnaround times with and without BCMID assistance.